

KAMYSHIN, Russian Federation

WMO: 343630

Lat: 50.083N

Lon: 45.367E

Elev: 119

StdP: 99.91

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-20.3	-26.4	0.3	-22.4	-23.7	0.4	-19.9	14.0	-6.9	12.3	-4.5	4.4	30	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	35.0	19.4	33.1	19.1	31.2	18.6	21.2	30.2	20.4	29.2	19.7	28.3	4.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.4	24.2	17.4	12.7	24.1	16.6	12.0	23.9	61.8	30.5	59.1	29.0	56.7	28.2	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.3	9.3	-25.6	37.8	3.7	2.1	-28.3	39.3	-30.4	40.5	-32.5	41.7	-35.1	43.3		
(5)			-25.0	22.4	3.9	1.3	-27.9	23.3	-30.2	24.0	-32.4	24.7	-35.2	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-7.9	-7.6	-1.1	9.5	16.9	21.9	24.4	23.1	16.5	8.0	0.4	-5.3	(6)
(7)		DBStd	12.84	6.33	6.72	5.22	5.03	4.90	4.13	3.86	4.60	4.53	4.73	5.34	5.98	(7)
(8)		HDD10.0	2333	556	494	345	68	6	0	0	0	4	96	290	474	(8)
(9)		HDD18.3	4204	814	727	603	267	87	13	2	11	88	321	539	733	(9)
(10)		CDD10.0	1716	0	0	1	54	221	356	447	406	198	33	1	0	(10)
(11)		CDD18.3	546	0	0	0	2	43	118	191	159	32	1	0	0	(11)
(12)		CDH23.3	5179	0	0	0	12	339	1076	1967	1572	211	3	0	0	(12)
(13)		CDH26.7	2096	0	0	0	1	97	418	857	672	51	0	0	0	(13)
(14)	Wind	WSAvg	4.7	4.8	5.1	5.1	4.8	4.6	4.5	4.2	4.3	4.6	4.7	5.0	(14)	
(15)	Precipitation	PrecAvg	378	30	24	22	22	34	43	36	33	35	24	36	36	(15)
(16)		PrecMax	595	68	92	75	64	95	101	122	133	89	74	80	85	(16)
(17)		PrecMin	191	2	3	1	2	1	12	6	0	0	0	5	12	(17)
(18)		PrecStd	94	16	19	15	14	28	26	27	28	27	19	21	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	4.2	13.9	25.0	31.7	36.7	37.6	37.2	31.1	22.3	13.0	6.1	(19)
(20)			MCWB	3.7	2.3	7.8	13.8	16.6	19.5	20.0	19.8	16.8	14.8	10.4	5.3	(20)
(21)		2%	DB	2.4	2.4	9.8	22.4	29.2	33.3	34.9	34.6	27.9	19.2	10.6	4.7	(21)
(22)			MCWB	1.7	1.3	5.8	13.4	16.6	19.0	19.8	19.0	16.1	12.4	8.7	3.9	(22)
(23)		5%	DB	1.2	1.5	7.0	19.5	26.9	30.9	33.1	32.7	25.8	16.7	8.8	3.0	(23)
(24)			MCWB	0.5	0.6	4.1	11.5	16.1	18.7	19.5	18.8	15.5	11.4	7.3	2.3	(24)
(25)		10%	DB	0.4	0.7	5.0	17.1	24.7	28.9	31.3	30.4	23.6	14.4	6.9	1.6	(25)
(26)			MCWB	-0.2	0.0	2.8	10.4	15.4	18.0	19.0	18.3	14.6	10.6	5.6	0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	2.7	8.6	15.6	19.2	21.7	22.2	22.3	18.8	16.1	11.0	5.6	(27)
(28)			MCDWB	4.2	3.8	12.4	21.8	27.4	31.2	32.0	30.4	25.4	20.4	12.4	5.9	(28)
(29)		2%	WB	1.9	1.6	6.1	13.9	17.9	20.6	21.2	20.8	17.6	14.0	9.2	4.0	(29)
(30)			MCDWB	2.2	2.2	9.0	20.1	26.8	29.0	30.7	29.8	24.0	17.3	10.5	4.6	(30)
(31)		5%	WB	0.7	0.8	4.4	12.5	16.9	19.6	20.6	20.0	16.6	12.3	7.4	2.3	(31)
(32)			MCDWB	1.1	1.4	6.7	18.6	25.0	28.1	29.6	28.8	23.6	15.4	8.6	2.8	(32)
(33)		10%	WB	-0.1	0.1	3.0	11.1	15.9	18.9	19.9	19.2	15.5	10.9	5.8	1.0	(33)
(34)			MCDWB	0.4	0.6	4.7	16.2	23.4	27.1	28.5	28.2	22.2	13.9	6.8	1.5	(34)
(35)	Mean Daily Temperature Range	MDBR	4.8	5.4	5.8	8.3	9.4	9.3	9.5	9.5	8.7	6.6	4.8	4.5	(35)	
(36)		5% DB	MCDBR	3.6	4.5	8.5	10.9	11.6	11.3	11.2	10.9	11.1	9.6	6.3	3.6	(36)
(37)			MCWBR	3.5	4.1	5.6	5.1	4.7	4.1	3.6	3.4	4.5	5.2	4.8	3.6	(37)
(38)		5% WB	MCDWB	3.5	4.3	7.9	10.0	10.6	10.0	10.0	9.8	9.5	8.4	5.8	3.5	(38)
(39)			MCWBR	3.5	4.2	5.5	5.2	5.1	4.2	3.8	3.9	4.8	5.4	5.1	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.345	0.376	0.440	0.409	0.396	0.411	0.426	0.382	0.355	0.332	0.314	(40)	
(41)		taud	2.111	2.026	2.114	2.082	2.270	2.348	2.307	2.242	2.341	2.360	2.291	2.152	(41)	
(42)		Ebn at Noon	684	757	813	800	849	863	842	806	807	760	675	624	(42)	
(43)		Edh at Noon	82	116	127	148	128	119	122	125	102	83	70	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.93	1.97	3.20	4.48	5.91	6.40	6.26	5.47	3.85	2.12	0.94	0.60	(44)	
(45)		RadStd	0.18	0.43	0.50	0.37	0.53	0.58	0.44	0.40	0.36	0.30	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page